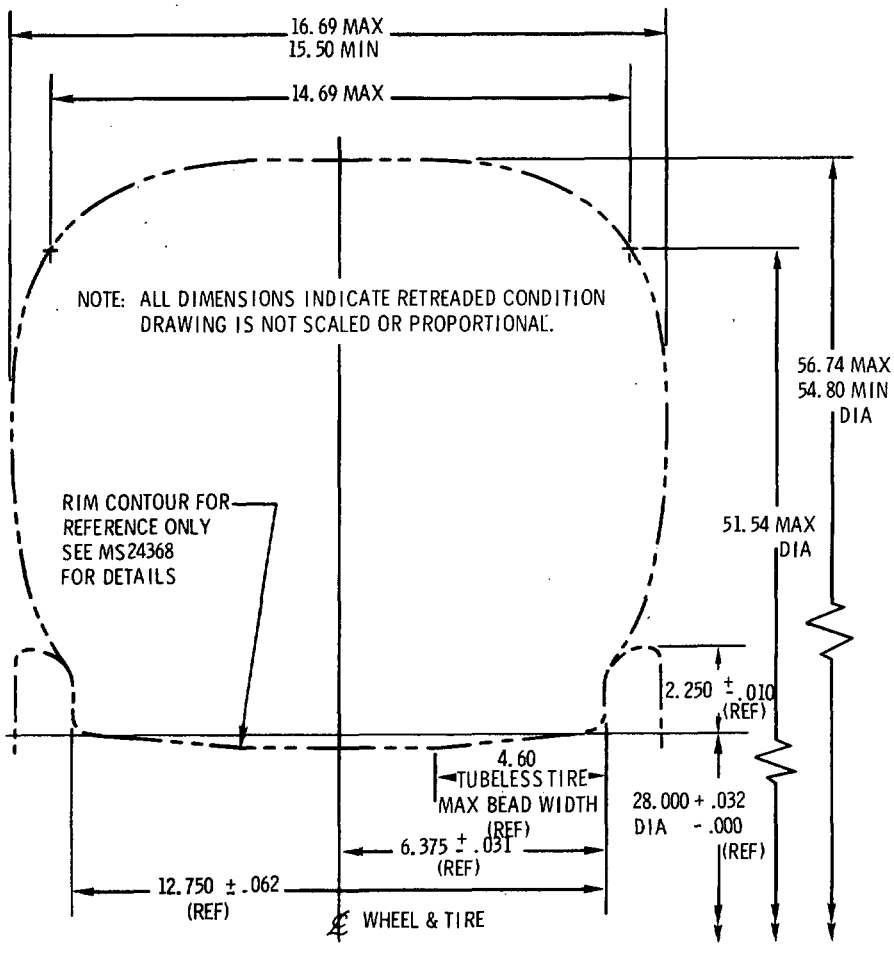
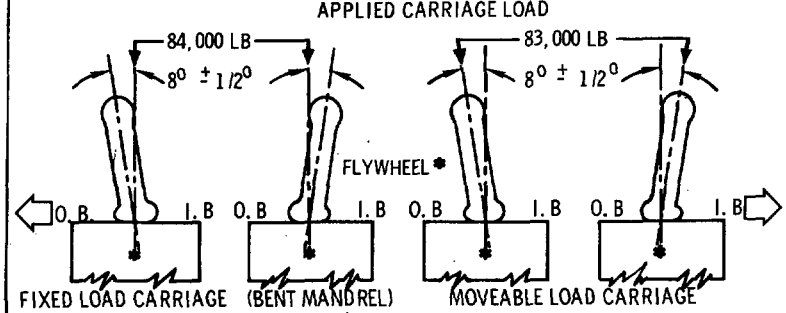
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NOTE: ALL DIMENSIONS ARE IN INCHES





TEST INFLATION PRESSURE
(38 PR)

FLYWHEEL DIA (IN)	TEST INFL PRES (PSI)
120	370
96	400

DETAIL C

FIGURE 1 Ⓐ FOR CHANGES SEE SHEETS 2 & 3

APPROVED 21 DEC 1970 REVISED Ⓐ 1 APRIL 1978

P.A. 99 Other Cust	TITLE TIRE - PNEUMATIC RETREAD, TYPE VII 56 X 16 / 38 PR 250 MPH	MILITARY STANDARD MS27813 (USAF)
PROCUREMENT SPECIFICATION MIL - R - 7726	SUPERSEDES: AF DWG NO. 62D31298	SHEET 1 OF 3

DD FORM 1 SEP 65 672-1 (Limited coordination) EDITION OF 1 SEP 57 MAY BE USED.

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GENERAL NOTES

1. ANY DEVIATION FROM GIVEN DIMENSIONS MUST BE AUTHORIZED BY WRITTEN APPROVAL OF THE PROCURING AGENCY. DIMENSIONS CALLED OUT ARE AFTER RETREADING.

2. REFERENCE TO ANY SPECIFICATION HEREIN SHOULD BE CONSTRUED AS REFERENCE TO THE ISSUE IN EFFECT ON THE DATE OF INVITATIONS FOR BIDS.

3. WHERE CONFLICT EXISTS BETWEEN THIS STANDARD AND ANY OTHER MILITARY SPECIFICATION, THE REQUIREMENTS OF THIS STANDARD SHALL APPLY.

4. THE QUALITY ASSURANCE SHALL BE IN ACCORDANCE WITH THE REQUIREMENTS OF MIL-R-7726.

RETREAD TIRE NOTES

THE RETREADED TIRE SHALL CONFORM TO THE APPLICABLE REQUIREMENTS OF MIL-R-7726 AND THE REQUIREMENTS LISTED BELOW:

SIZE	RETR TIRE PR	INFLATION PRESSURE PSI(RATED)+	BEAD WIDTH (MAX) IN	WEIGHT LBS MAX	STATIC UNBALANCE IN OZ MAX	TREAD	MOLD SKID DEPTH	MAX STATIC LOAD LBS
36X16	38	315	4.60+	379	90	RIB (SEE NOTE 4)	0.35 IN (MIN)	76,000

+ MAXIMUM BEAD TOE FLASH OF 3/8 INCH NOT INCLUDED.

1. THREE (3) CONSECUTIVE TIRES ARE TO SUCCESSFULLY COMPLETE THE DYNAMOMETER TESTING: THE FIRST TIRE SHALL BE TESTED TO THE ENTIRE TEST SPECTRUM UNDER TIRE NOTE 2: THE REMAINING TWO (2) TIRES ARE TO BE TESTED TO TEST CONDITIONS "A" (HIGH SPEED TAKE-OFF) AND "B" (LOW SPEED TAKE-OFF).

2. THE TIRE SHALL SATISFACTORILY WITHSTAND 10 CYCLES OF TEST "A", 30 CYCLES OF TEST "B", 40 CYCLES OF TEST "C", 8 CYCLES OF TEST "D", AND 60 CYCLES OF TEST "E" (30 CYCLES E_1 - 30 CYCLES OF E_2), AS NOTED BELOW WITHOUT EVIDENCE OF FAILURE. TESTS "D" AND "E" SHALL BE PERFORMED LAST WITH TEST "D" PRECEDING "E". THE SEQUENCE OF TESTING FOR TESTS "A", "B" AND "C" IS OPTIONAL.

A. HIGH SPEED TAKE-OFF - THE TIRE SHALL BE TAXIED ON THE FLYWHEEL FOR 14,000 FT AT STATIC LOAD RATING AT 30 MPH. STOP THE FLYWHEEL UNDER FULL LOAD AND THEN ACCELERATE IT (SIMULATING TAKE-OFF) AT AN AVERAGE RATE OF 5 FT/SEC/SEC TO 250 MPH. THE TIRE SHALL BE UNLANDED AFTER A TAKE-OFF ROLL DISTANCE OF 13,500 ± 200 FEET HAS BEEN COVERED. THE LOAD SHALL BE MAINTAINED AT STATIC LOAD RATING (POUNDS) FOR THE FIRST 18 TO 20 SECONDS DURING TAKE-OFF ROLL. THE LOAD SHALL THEN BE DECREASED LINEARLY WITH TIME IN 65 TO 68 SECONDS AFTER ACCELERATION BEGINS TO 40,000 LBS AND FURTHER REDUCED LINEARLY WITH TIME TO ZERO LOAD AT THE END OF THE TAKE-OFF ROLL.

B. LOW SPEED TAKE-OFF - THE TIRE SHALL BE TAXIED ON THE FLYWHEEL FOR 14,000 FEET AT STATIC LOAD RATING AND 30 MPH. STOP THE FLYWHEEL UNDER FULL LOAD AND THEN ACCELERATE IT (SIMULATING TAKE-OFF) AT AN AVERAGE RATE OF 4 FT/SEC/SEC TO 214 MPH. THE TIRE SHALL BE UNLANDED AFTER A TAKE OFF ROLL DISTANCE OF 12,300 ± 200 FT HAS BEEN COVERED. THE LOAD SHALL BE MAINTAINED AT STATIC LOAD RATING FOR THE FIRST 26 TO 28 SECONDS DURING THE TAKE-OFF ROLL. THE LOAD SHALL THEN BE DECREASED LINEARLY WITH TIME IN 64 TO 66 SECONDS AFTER ACCELERATION BEGINS TO 44,000 LBS AND FURTHER REDUCED LINEARLY WITH TIME TO ZERO LOAD AT THE END OF THE TAKE-OFF ROLL.

C. LANDING - THE TIRE SHALL BE LANDED AGAINST THE FLYWHEEL ROTATING AT A PERIPHERAL SPEED OF 165 MPH. THE AVERAGE DECELERATION RATE SHALL BE 6.3 FT/SEC/SEC FROM 165 MPH TO 30 MPH. THE LOAD SHALL BE INCREASED LINEARLY DURING THE INITIAL 8 to 9 SECONDS AFTER LANDING TO 36,000LBS, THEN FURTHER INCREASED TO 45,000LBS IN 30 TO 33 SECONDS AFTER LANDING AND A ROLL DISTANCE OF 4500±200 FEET HAS BEEN COVERED. THE TIRES SHALL THEN BE TAXIED AT A CONSTANT SPEED OF 30MPH FOR 14,000FT AT 45,000LBS LOAD.

D. TAXI-ROLL - THE TIRE SHALL BE LANDED AGAINST A FLYWHEEL ROTATING AT A PERIPHERAL SPEED OF 30 MPH AND TAXIED FOR 35,000 FT AT STATIC LOAD RATING. THE TIRE SHALL BE ALLOWED TO COOL TO 120° ± 10°F BEFORE REPEATING THE CYCLE.

E. TAXI-CAMBER - CONDUCT CAMBER TESTS IN ACCORDANCE WITH THE "MOVABLE LOAD CARRIAGE" PROCEDURE (DETAIL A,) IF THIS CAPABILITY DOES NOT EXIST, AN ACCEPTABLE ALTERNATE IS THE FIXED LOAD CARRIAGE "BENT MANDREL" PROCEDURE (DETAIL B).

E_1 TAXI-OUTBOARD CAMBER:

1. ACCELERATE FLYWHEEL TO 15 MPH AND MAINTAIN THIS SPEED THROUGHOUT THE ENTIRE TEST.
2. LAND THE TIRE AGAINST THE FLYWHEEL IN A POSITION OF 8°± ½° OUTBOARD CAMBER AND LOAD AS FOLLOWS:
   A. INCREASE THE CARRIAGE LOAD LINEARLY TO 50,000 LBS DURING THE FIRST 2 SECONDS.
   B. FURTHER INCREASE THE LOAD TO 83,000 LBS WITHIN 3.5 TO 4 SECONDS AFTER LANDING PER DETAIL A, OR 84,000 LBS PER DETAIL B.
   C. MAINTAIN THIS LOAD UNTIL A DISTANCE OF 3,350 FEET HAS BEEN COVERED.
3. UNLAND THE TEST TIRE.

E_2 TAXI-INBOARD CAMBER

1. SAME AS E_1 EXCEPT LAND THE TEST WHEEL IN A POSITION OF 8° ± ½° INBOARD CAMBER.

3. TEST INFLATION PRESSURE: THE TIRE INFLATION PRESSURE USED FOR ALL CYCLES SHALL BE IN ACCORDANCE WITH DETAIL C FOR THE RESPECTIVE DIAMETER FLYWHEEL USED TO CONDUCT THE CYCLE(S). THE TIRE OUTSIDE DIAMETER, ROLLING RADIUS, AND DEFLECTION ON THE FLYWHEEL AT RATED LOAD AND TEST INFLATION PRESSURE FOR THE CORRESPONDING FLYWHEEL DIAMETER SHALL BE RECORDED AND STATED IN THE QUALIFICATION TEST REPORT AFTER A MINIMUM OF 12 HOURS STATIC GROWTH AT THE RATED INFLATION PRESSURE.

APPROVED 21 DEC 1970 / REVISED (A) FOR CHANGES SEE SHEETS 2 & 3

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P.A. 99 Other Cust	TITLE: TIRE - PNEUMATIC RETREAD, TYPE VII 56 X 16 / 38 PR 250 MPH	MILITARY STANDARD MS27813 (USAF)
PROCUREMENT SPECIFICATION MIL-R-7726	SUPERSEDES: AF. DWG. NO. 62D31298	SHEET 2 OF 3

DD FORM 1 SEP 63 672-1 (Limited coordination) EDITION OF 1 SEP 57 MAY BE USED.

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4. THE TREAD PATTERN SHALL HAVE A MINIMUM OF FOUR GROOVES. ALL GROOVES SHALL BE A MINIMUM OF THE SKID DEPTH SPECIFIED ABOVE. IF THE TREAD PATTERN HAS FIVE OR MORE GROOVES THE TWO OUTER GROOVES MAY BE OF A PARTIAL SKID DEPTH (MINIMUM OF 80% OF THE ACTUAL MOLD SKID DEPTH). A MINIMUM OF FOUR GROOVES SHALL BE WITHIN THE CONTACT AREA TAKEN AT 60% OF THE MAXIMUM STATIC LOAD AT RATED INFLATION PRESSURE. IF THE TREAD PATTERN HAS A CENTER RIB, THE MAXIMUM CENTER RIB WIDTH OF THE UNINFLATED TIRE SHALL NOT EXCEED 3.0 INCHES. THE MOLD GROOVE WIDTH SHALL BE A MINIMUM OF 0.25 INCH WHEN MEASURED AT 50% OF THE MOLD GROOVE DEPTH. ALL OF THE ABOVE SHALL BE REPORTED IN THE QUALIFICATION TEST REPORT.

5. THE TUBELESS TIRE MAY BE CONSTRUCTED WITH A FABRIC REINFORCED TREAD FEATURE FOR HIGH SPEED OPERATION AND MAXIMUM RELIABILITY AND RETENTION OF THE TREAD TO THE CARCASS.

APPROVED 21 DEC 1970 REVISED (A) FOR CHANGES SEE SHEETS 2 & 3

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